Arista Access Node YEL-ACC-32 QoS Congestion Event Resolution SOP

Node: YEL-ACC-32 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_9359

Description

On Arista Access nodes, specifically YEL-ACC-32 in the Yelahanka Sub-station, QoS congestion events can occur due to microbursts, interface tail drops, or QoS queue starvation. This phenomenon is largely attributed to the interaction between Class of Service (CoS) queues, packet scheduling algorithms, and buffer allocation mechanisms. The Arista Access node employs a hierarchical CoS architecture, with multiple queues per interface (e.g., queue 0-7) designed to handle diverse packet types (e.g., voice, video, best-effort). When these queues become congested, packets are either dropped or delayed, resulting in QoS degradation. Furthermore, the node's buffer allocation strategy, coupled with the packet scheduling algorithm, can exacerbate queue congestion during microbursts. This SOP aims to troubleshoot and resolve QoS congestion events on YEL-ACC-32 using Arista-specific CLI commands.

Pre-requisite Checks

- Ensure remote hands are dispatched to Yelahanka Sub-station for physical access to the node.

- Verify that all redundant links are functioning properly.

- Confirm that the node's power supply is stable and within operational limits.

- Check the node's environmental conditions, such as temperature and humidity, to ensure they are within recommended ranges.

Potential Root Causes

Cause	Probability
Microburst-induced queue congestion due to misconfigured packet scheduling algorithms.	High
Insufficient buffer allocation for CoS queues.	Medium
QoS queue starvation due to incorrect queue priority settings.	Low

Diagnostic Commands & Expected Outputs

```
Command: show hardware internal qos queue counters, Expected Output: Queue counters to determine which queues are congested., Description: This command provides detailed queue counters, including packets dropped, delayed, and sent, to help identify which queues are experiencing congestion. By analyzing these counters, we can determine which queues are being impacted and isolate potential root causes.

Command: show hardware internal qos buffer allocation, Expected Output: Buffer allocation details to assess available resources.,
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```
Description: This command displays buffer allocation details, including the amount of available buffer space and allocation strategies. By analyzing this output, we can assess whether the buffer allocation is adequate to handle the traffic load and identify potential issues that may be contributing to congestion.

Command: show hardware internal qos packet scheduling, Expected Output: Packet scheduling details to evaluate algorithmic performance., Description: This command provides detailed packet scheduling information, including the scheduling algorithm in use, packet priorities, and queue servicing times. By analyzing this output, we can evaluate the performance of the packet scheduling algorithm and identify potential issues that may be contributing to congestion.
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Resolution Steps

Step 1: Step: Adjust packet scheduling algorithm, Command: config hardware internal qos packet scheduling algorithm , Description: Adjust the packet scheduling algorithm to optimize performance and prevent congestion. Select an algorithm that balances packet priority and servicing time to ensure optimal QoS performance.

Step 2: Step: Reconfigure buffer allocation, Command: config hardware internal qos buffer allocation , Description: Reconfigure buffer allocation to ensure sufficient resources are available for CoS queues. Choose an allocation strategy that balances buffer space and allocation efficiency to prevent congestion.

Step 3: Step: Reorder CoS queues, Command: config hardware internal qos queue reorder , Description: Reorder CoS queues to ensure priority packets are serviced first. This step is crucial in preventing QoS queue starvation and ensuring optimal performance.

Post-Check Verification

```
Command: show hardware internal qos queue counters, Expected Output: Queue counters indicating improved performance., Description: Run this command to verify that the changes have improved queue performance and reduced congestion.

Command: show hardware internal qos buffer allocation, Expected Output: Buffer allocation details indicating sufficient resources., Description: Verify that buffer allocation has been adjusted to provide sufficient resources for CoS queues.

Command: show hardware internal qos packet scheduling, Expected Output: Packet scheduling details indicating optimal performance., Description: Confirm that the packet scheduling algorithm has been adjusted to optimize performance and prevent congestion.
```

Roll-back Procedures

- Step: Restore original packet scheduling algorithm, Command: config hardware internal qos packet scheduling algorithm , Description: Restore the original packet scheduling algorithm to prevent performance degradation.

- Step: Restore original buffer allocation, Command: config hardware internal qos buffer allocation , Description: Restore the original buffer allocation strategy to prevent resource exhaustion.

- Step: Restore original CoS queue order, Command: config hardware internal qos queue reorder , Description: Restore the original CoS queue order to prevent QoS queue starvation.